Project Title -Seasonal Agriculture Performance Analysis

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PROBLEM STATEMENT

- (Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.

Project Description

The project analyzes a seasonal agriculture dataset
containing 4,000 farm
records and 28 attributes. The analysis focuses on crop
performance, seasonal
trends, state-wise performance, irrigation methods, yield,
production, revenue,
cost, profit, water usage and disease/pest risk. Data
visualization is used to
convert the raw dataset into meaningful insights that can
support better
agricultural decision-making

WHO ARE THE END USERS?

- Farmers
- Agricultural consultants
- Agriculture researchers
- Government agriculture departments
- Irrigation planners
- Agricultural businesses

Technology Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook / Google Colab
- CSV Datase

RESULTS 1 - Dataset & Data Analysis

Seasonal_Agricultural_Performance_Data_Analytics.ipynb

```
FilePath = "/content/seasonal_agriculture_performance_dataset.csv"
df = pd.read_csv(FilePath)
print("dataset loadesd sucessfully")
```

```
dataset loadesd sucessfully
```

```
print("number of rows:",df.shape[0])
print("number of columns:",df.shape[1])
```

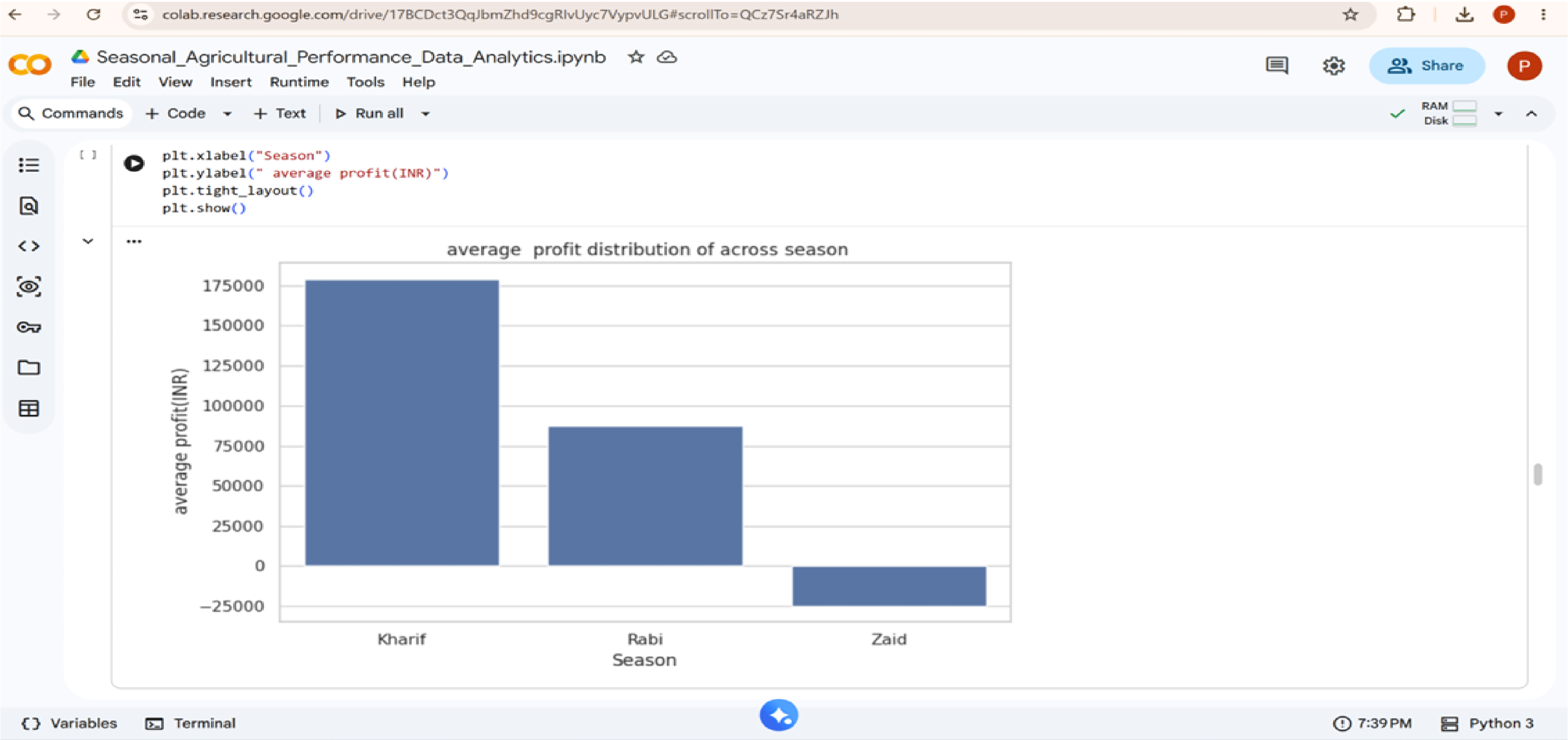
```
number of rows: 4000
number of columns: 28
```

```
df.head()
```

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	Soil_pH	Soil_Moisture_pct
0	SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif	0.53	486.40	24.30	69.10	5.80	7.14	32.80
1	SF10002	Maharashtra	Nalgonda	Maize	Kharif	6.53	855.40	25.70	78.40	5.10	5.20	32.00
2	SF10003	Telangana	Warangal	Pulses	Rabi	4.86	455.60	23.90	56.40	10.30	5.82	8.10
3	SF10004	Telangana	Indore	Rice	Kharif	4.50	753.20	29.70	65.20	6.30	5.98	23.70
4	SF10005	Karnataka	Ludhiana	Maize	Zaid	4.21	101.60	34.10	55.20	6.50	6.90	29.30

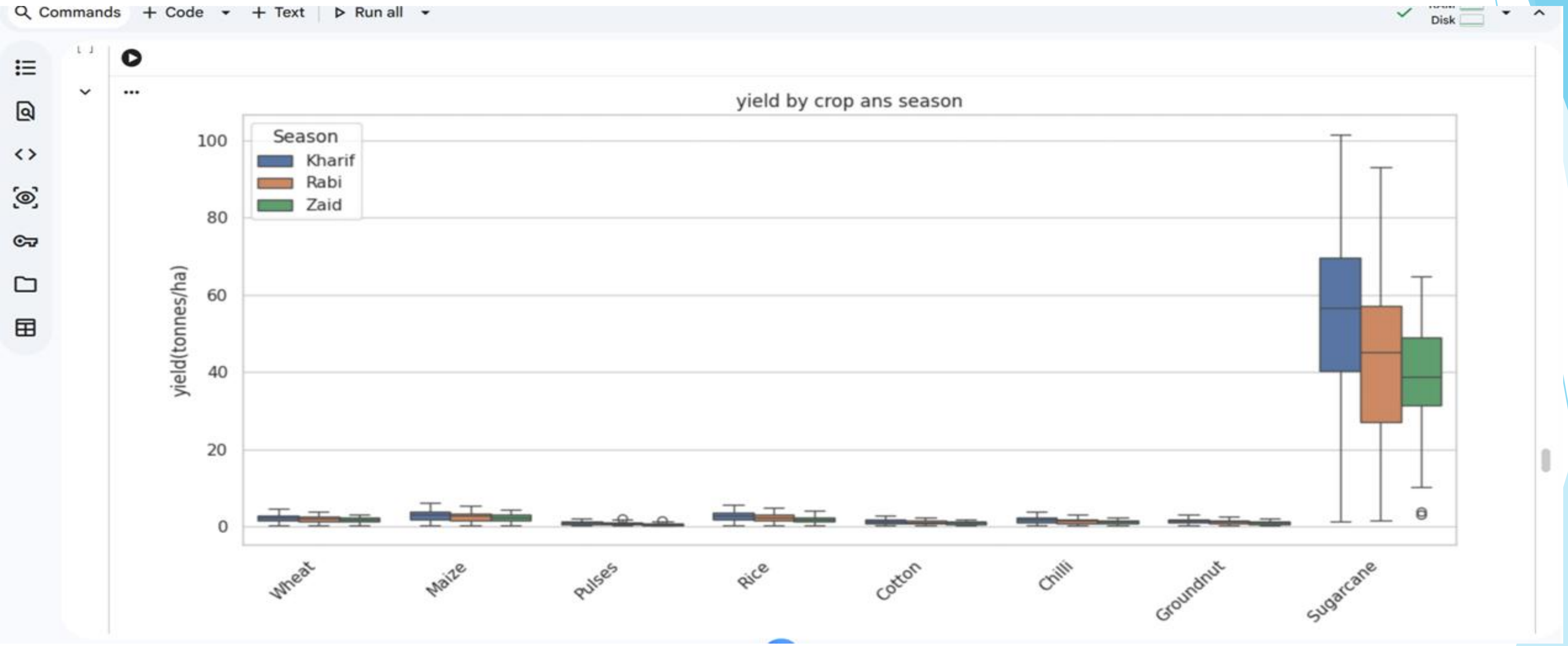
[The dataset contains 4,000 farm records and 28 agricultural attributes. The analysis includes crop, season, rainfall, temperature, soil conditions, irrigation, yield, production, cost, revenue and profit.]

RESULTS 2:Seasonal Performance



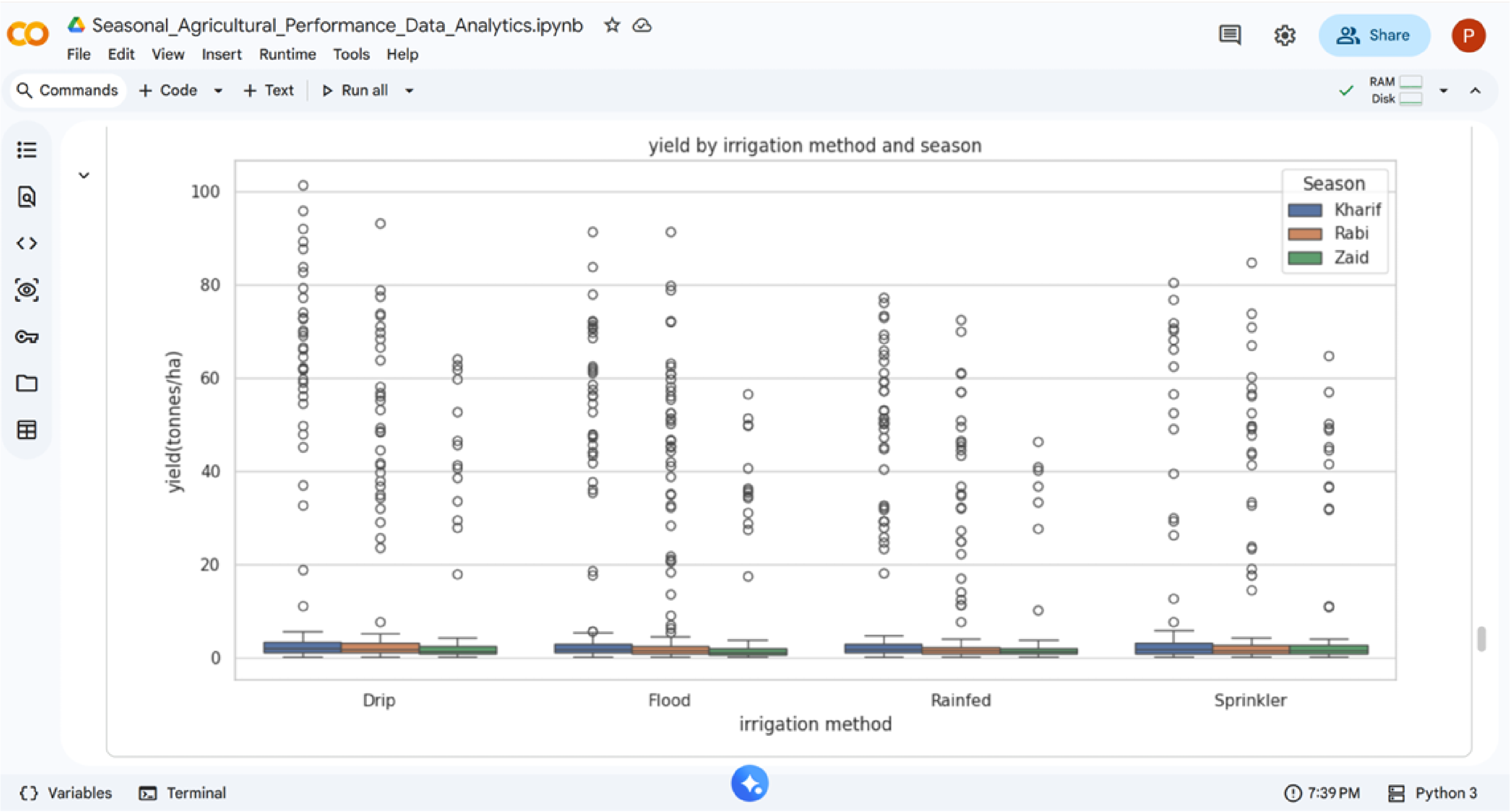
[Seasonal analysis was performed to compare agricultural profitability across Kharif, Rabi and Zaid seasons. The comparison helps identify which season provides better overall farm performance.]

RESULTS 3: Crop & Season Analysis



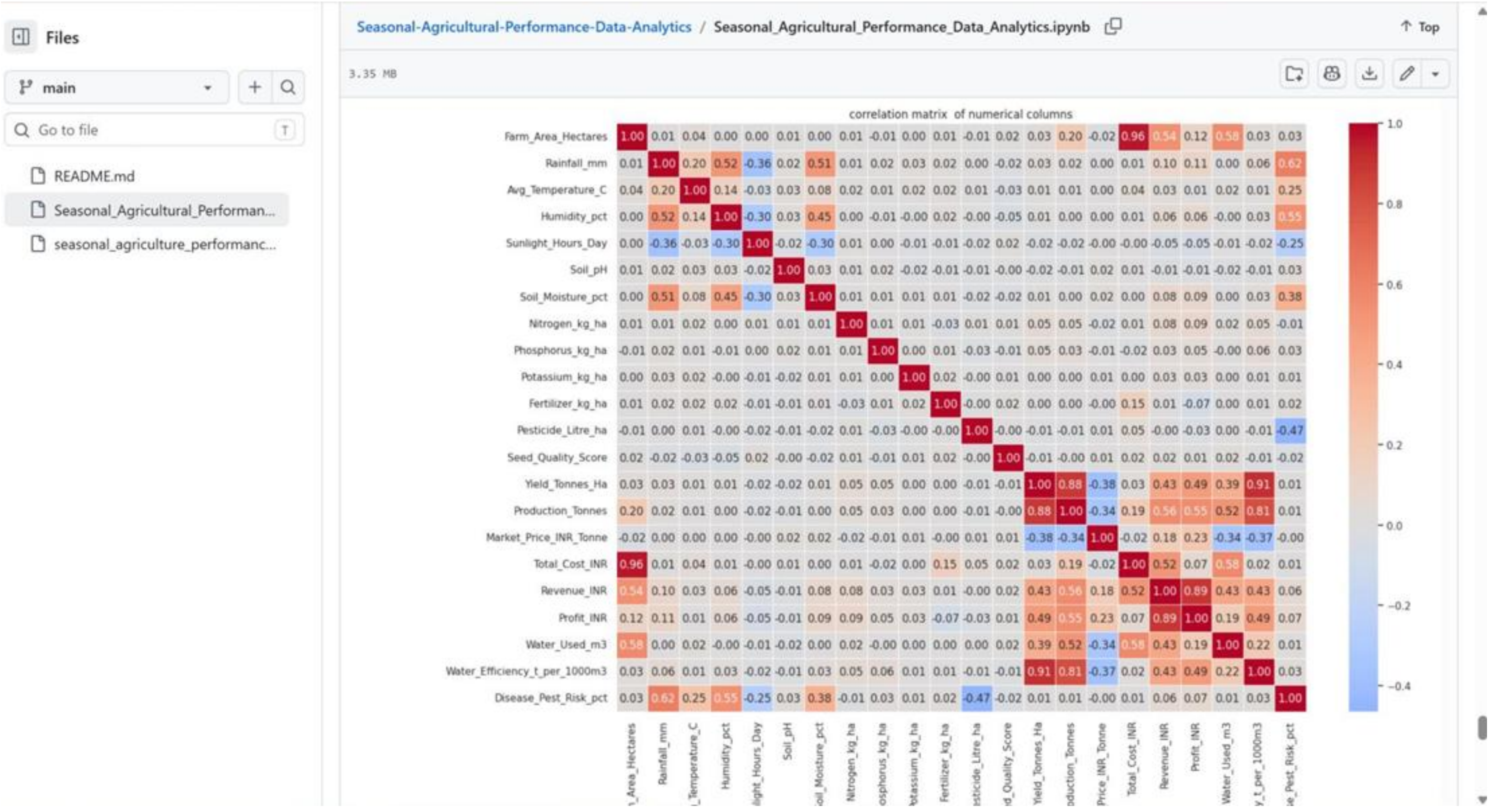
Crop-wise and season-wise analysis was performed to understand how crop productivity changes across different agricultural seasons. This comparison helps identify better-performing crop-season combinations.]

RESULTS 4: Irrigation Analysis



[[Irrigation methods were compared based on crop yield across different seasons. The analysis helps understand the relationship between irrigation practices and agricultural productivity.]

RESULTS 5 :Correlation Analysis



[Correlation analysis was performed on the numerical agricultural variables to identify relationships between rainfall, temperature, soil parameters, yield, production, profit and water usage. The heatmap provides a quick overview of strong and weak relationships within 10 the dataset.

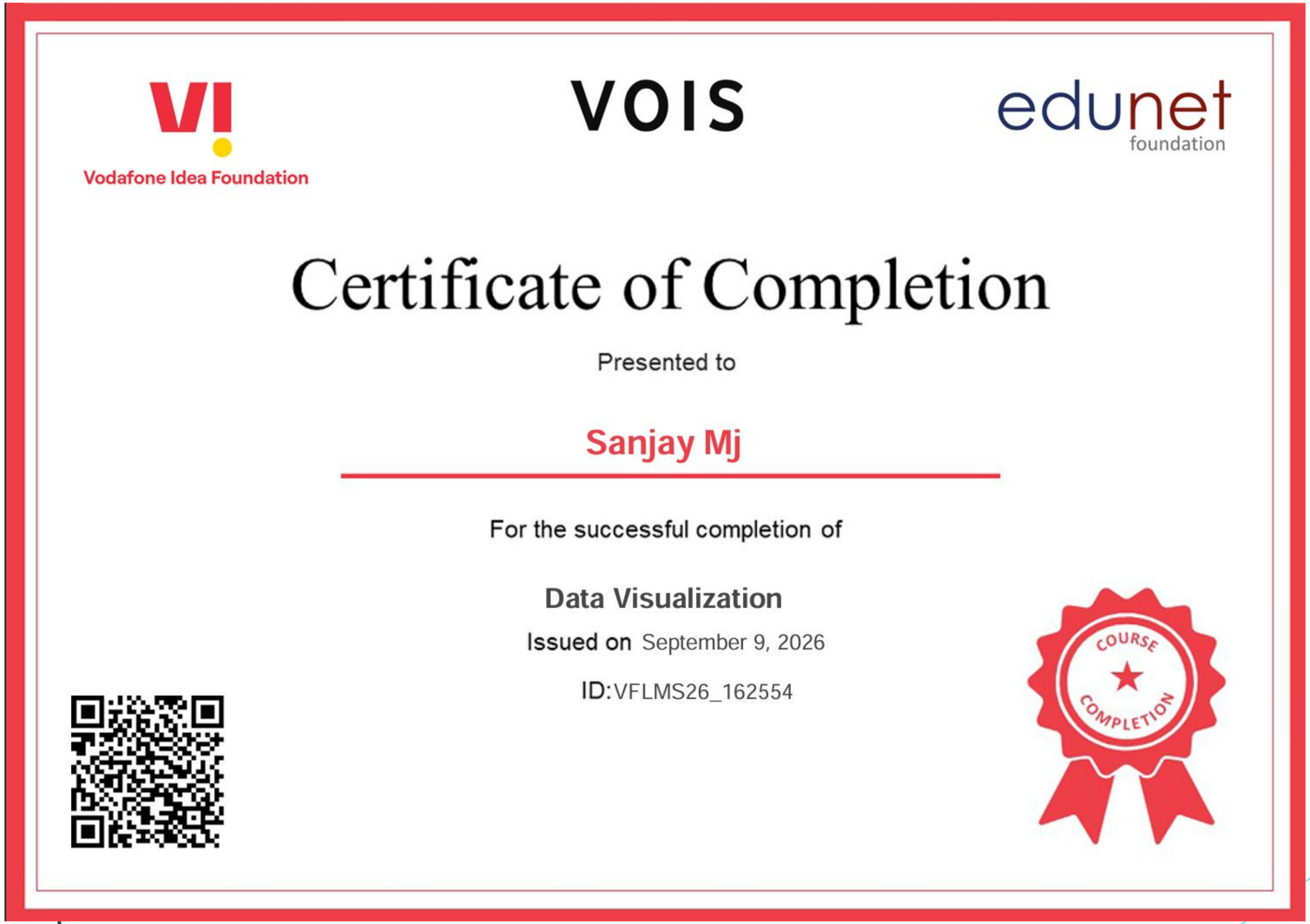
Future scope

- Build an interactive Power BI dashboard.
- Add machine-learning based yield prediction.
- Develop crop recommendation systems.
- Predict agricultural profit before cultivation.
- Add weather forecasting data.
- Provide irrigation and resource optimization recommendations

GitHub Link

https://github.com/sanjaymj1/Seasonal-Agricultural-Performance-Data-Analytics.git

VOIS Course completion certificate - Course name- Data Visualization



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Sanjay Mj

For the successful completion of

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Issued on September 9, 2026

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COURSE COMPLETION

Thank you